

COMP 4332 / RMBI 4310

Big Data Mining (Spring 2026)

Project 2: TabPFN

TA: Yisen Gao (ygaodi@connect.ust.hk)

Tabular Data

Very common in real-world applications

Industry
Business
Medical
...

Rows = observations

Tabular Data

columns = attributes for those observations

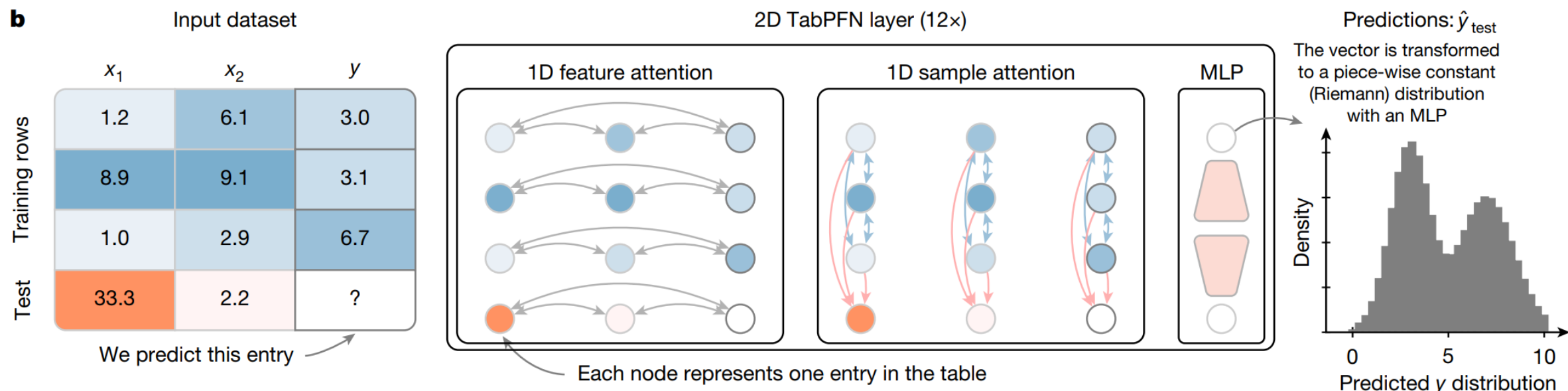
Column 1	Column 2	Column 3	Column 4	Column 5

TabPFN

In TabPFN v2, tokens are individual cells (a single feature value or label), not entire rows.

During **Sample Attention**, the sequence consists of all cells *within the same feature column*. Tokens attend only to other cells within the same column. **The Train Samples:** Features and labels are encoded as separate, independent cell tokens.

This allows a test cell to find training cells with similar values, essentially performing a sophisticated, learned 1D k-Nearest Neighbors or Kernel Regression per feature.



Tabular Regression/Classification

- Regression: Predict the value; evaluate with RMSE.
- Classification: Predict the class; evaluate with Accuracy.

	A	B	C	D	E	F	G	H	I	J	K	L
1	fixed acid	volatile acid	citric acid	residual sugar	chlorides	free sulfur	total sulfur	density	pH	sulphates	alcohol	quality
2	7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5
3	7.8	0.88	0	2.6	0.098	25	67	0.9968	3.2	0.68	9.8	5
4	7.8	0.76	0.04	2.3	0.092	15	54	0.997	3.26	0.65	9.8	5
5	11.2	0.28	0.56	1.9	0.075	17	60	0.998	3.16	0.58	9.8	6
6	7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5
7	7.4	0.66	0	1.8	0.075	13	40	0.9978	3.51	0.56	9.4	5
8	7.9	0.6	0.06	1.6	0.069	15	59	0.9964	3.3	0.46	9.4	5
9	7.3	0.65	0	1.2	0.065	15	21	0.9946	3.39	0.47	10	7
10	7.8	0.58	0.02	2	0.073	9	18	0.9968	3.36	0.57	9.5	7
11	7.5	0.5	0.36	6.1	0.071	17	102	0.9978	3.35	0.8	10.5	5
12	6.7	0.58	0.08	1.8	0.097	15	65	0.9959	3.28	0.54	9.2	5
13	7.5	0.5	0.36	6.1	0.071	17	102	0.9978	3.35	0.8	10.5	5
14	5.6	0.615	0	1.6	0.089	16	59	0.9943	3.58	0.52	9.9	5
15	7.8	0.61	0.29	1.6	0.114	9	29	0.9974	3.26	1.56	9.1	5
16	8.9	0.62	0.18	3.8	0.176	52	145	0.9986	3.16	0.88	9.2	5
17	8.9	0.62	0.19	3.9	0.17	51	148	0.9986	3.17	0.93	9.2	5
18	8.5	0.28	0.56	1.8	0.092	35	103	0.9969	3.3	0.75	10.5	7
19	8.1	0.56	0.28	1.7	0.368	16	56	0.9968	3.11	1.28	9.3	5
20	7.4	0.59	0.08	4.4	0.086	6	29	0.9974	3.38	0.5	9	4

Dataset

- ▣ Ten tabular data with 5 for regression task and 5 for classification task.
 - Each one have train.csv, test.csv and result.csv
 - **The last column** is your target prediction
 - You can see the detailed data in the 'data/reg' and 'data/cls' file

Method

- ▣ You can use tabpfn or its variants like Tarte, Limix.
 - Note that you do not need to train the model
 - Note that there may be a limit on the maximum number of lines allowed in the context.

Evaluation

- **RMSE** for regression dataset on **test data**
- **Accuracy** for classification dataset on **test data**
- You should generate the 'predict.csv' for the target column
- Use the 'evaluate.py' to get your performance.

Submission

- Report (1~2 pages), including the motivation of your model choice, the choice of the context (if there are too many lines), and the comparison of model performances.
- Code
- predict.csv
- DDL: **April 26, 2026**
- Submission: Each **team leader** is required to submit the **groupNo.zip** file that contains the report, results and your team's code on canvas.
- We will check your report with your code and the Accuracy/RMSE scores. You will be graded based on your testing set performance and your report.

Grading Rule

- The baseline for each dataset is listed below. For each dataset you outperform, you will earn 10% of the total score.
- We will also adjust your final grade based on your report.

Table 1. Baseline Performance on Classification Dataset

dataset	accuracy
1	0.86
2	0.70
3	0.64
4	0.68
5	0.70

Table 2. Baseline Performance on Regression Dataset

dataset	rmse
1	2.60
2	3.60
3	0.70
4	0.03
5	0.21

Thank You